Introduction to Machine Learning — Final Project

Konstantinos Papafilippou[1], Duru Yigit[2], Federica Gatti[3]
Group RR10 , prof. Ricardo Ribeiro

This report presents the outcomes and methodologies of the project for Introduction to Machine Learning — 2025/2026. It outlines the steps followed according to the CRISP-DM framework and includes the exercises required to complete the project.
The project has been developed principally in group meetings, so that the contribution of each group member is present in every step of the project.

1 Business understanding

The project is based on energy consumption data of Municipality of Maia. Energy data set is composed by structure (CPE) label, timestamp, Consumption Data (Data de Consumo), Active Power (PotAct), Reactive Inductive Power (PotReactIndut) and Reactive Capacity Power (PotReactCapac). As announced in Moodle, Dados de Consumo column will be neglected during the following study.
These data represent the historical energy consumption from January 2023 to May 2025. They come from different types of CPE, each with distinct features. The aim of the project is to characterize and describe these CPE types in order to predict their future behavior. To achieve this, the Data Preparation and Data Understanding steps are crucial for identifying the most relevant features. Based on these features, a clustering operation was performed. These clusters will then be used for prediction purposes.

2 Data Understanding and Preparation

Data Preparation - Missing values and Basic statistics As a first step, a main Data Preparation phase was carried out. The data were extracted from the storage system and processed using the Pandas library. Consumption data was removed from the dataset. The timestamp column was parsed into the datetime64 format (figure 1).

From the first five samples (figure 2) it is possible to observe that each row represents an energy reading from a CPE, with measurements taken every 15 minutes. Moreover, some data present "Null" value, then missing ratio of each columns has been performed (figure 3).

[1]kpuso@iscte-iul.pt , 137075
[2]dytuu@iscte-iul.pt , 136603
[3]fgiae1@iscte-iul.pt , 134615

```
Data columns (total 5 columns):
 #   Column         Non-Null Count    Dtype
---  ------         --------------    -----
 0   CPE            1000000 non-null  category
 1   tstamp         1000000 non-null  datetime64[ns]
 2   PotActiva      1000000 non-null  float32
 3   PotReactIndut  991268 non-null   float32
 4   PotReactCapac  991268 non-null   float32
dtypes: category(1), datetime64[ns](1), float32(3)
memory usage: 20.0 MB
```

Figura 1: Data columns, Non-null counts and Dtype.

```
                    CPE              tstamp  PotActiva  PotReactIndut  \
0  PT0002000089069203YG 2022-12-31 00:15:00      0.296            NaN
1  PT0002000089069203YG 2022-12-31 00:30:00      0.292            NaN
2  PT0002000089069203YG 2022-12-31 00:45:00      0.284            NaN
3  PT0002000089069203YG 2022-12-31 01:00:00      0.280            NaN
4  PT0002000089069203YG 2022-12-31 01:15:00      0.272            NaN

   PotReactCapac
0            NaN
1            NaN
2            NaN
3            NaN
4            NaN
```

Figura 2: First five samples structure

Records without timestamps were removed, and the data were sorted by CPE and timestamp. Basic statistics for each numeric column has been calculated in figure 4. A large dataset of 3,668,090 non-null values is present, where mean value of inductive reactive power is higher than capacitive reactive power. Standard deviation values show that inductive power varies more widely than capacitive power. 50% percentile (median) and 75% one are very low, meaning most of the values are zero or near to zero, with occasional large values. Outliers are depicted by 95 and 99 percentile. Inductive power seems more variable and might be more informative for clustering.
A check on time frequency revealed missing data (figure 5), which required a subsequent data cleaning assessment. Duplicates in CPE and timestamps rows are removed.

```
Missing ratio:
PotReactCapac    0.382367
PotReactIndut    0.382367
PotActiva        0.000000
CPE              0.000000
dtype: float64
```

Figura 3: Missing ratio per column

	PotActiva	PotReactIndut	PotReactCapac
count	5.940182e+06	3.668710e+06	3.668710e+06
mean	8.093968e+00	1.715323e+00	6.525602e-01
std	2.229668e+01	3.364733e+00	1.667429e+00
min	0.000000e+00	0.000000e+00	0.000000e+00
25%	1.200000e-01	0.000000e+00	0.000000e+00
50%	2.000000e+00	0.000000e+00	0.000000e+00
75%	7.000000e+00	2.000000e+00	1.000000e+00
95%	3.500000e+01	9.000000e+00	3.000000e+00
99%	7.500000e+01	1.600000e+01	8.000000e+00
max	5.740000e+02	1.770000e+02	1.240000e+02

Figura 4: Basic statistics for each value

tstamp	count
0 days 00:15:00	78770
0 days 00:30:00	14
0 days 00:00:00	8
0 days 01:15:00	3
31 days 00:30:00	1

dtype: int64

Figura 5: Time frequency check

Each CPE is handled individually. For each of them, when a row is completely missing (i.e. a measurement has not been performed every 15 minutes), it is created and filled the numerical columns by computing a temporal linear interpolation.

Data Understanding The distributions of Active Power, Reactive Inductive Power, and Reactive Capacitive Power are shown in the figure 6, illustrating the range of mean values and the presence of outliers.

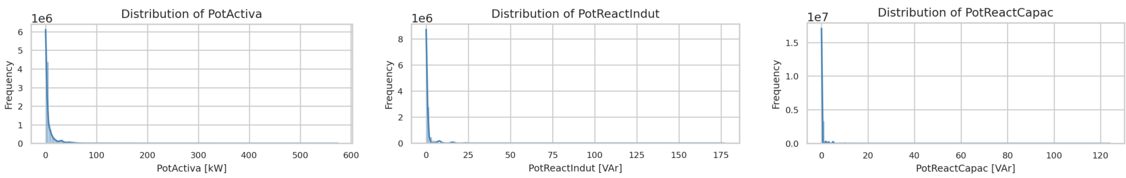


Figura 6: Distribution plots of Active Power, Reactive Inductive Power, and Reactive Capacitive Power.

The same information can be represented using a boxplot (Figure 7), which provides a more detailed view of the distribution and allows for a clearer identification of outliers.

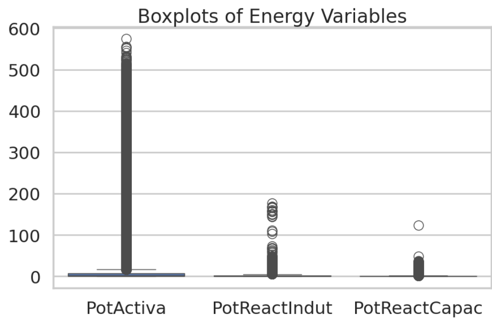


Figura 7: Boxplots of energy values

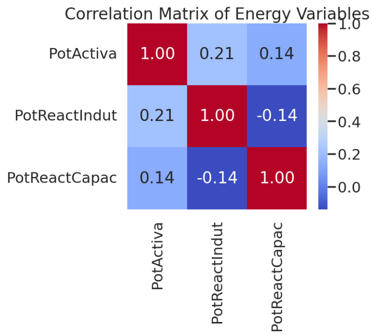


Figura 8: Correlation matrix between energy values

Correlation matrix between the three Energy values is depicted in 8, where they are only loosely related linearly. The slight negative link between inductive and capacitive reactive components is due to the fact they represent opposite types of reactive power, and the small magnitude suggests the dataset often has one dominant regime at a time.

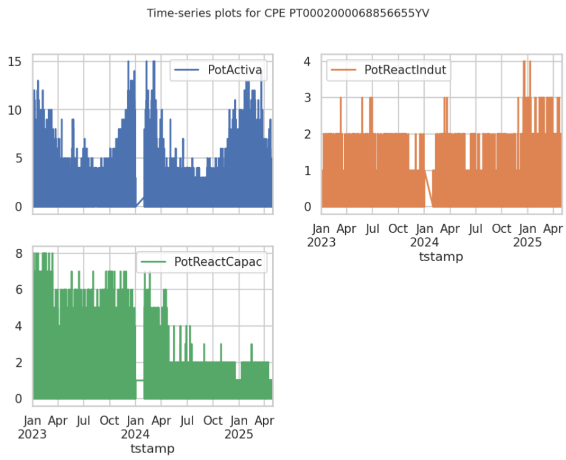


Figura 9: Time-series visualization for one CPE

Figure 9 presents a time-series visualization for a sample CPE, clearly illustrating the missing measurement month that was previously highlighted in Figure 5.

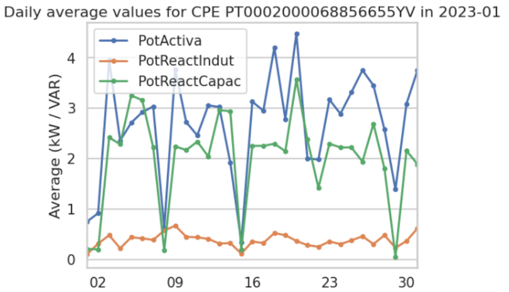
2.1 Features selection

Initial brainstorming: single CPE study The selection of meaningful features for characterizing the dataset is now being addressed. Timestamps are aggregated at both daily and monthly levels to analyze the following aspects:

- Daily profiles within each month

- Hourly profiles within a day

By examining one selected CPE, it becomes evident that valuable insights can be extracted from these profiles. For example, Figure 10 shows that the CPE does not consume energy during weekends (likely because the facility is closed) and that energy consumption is mainly concentrated between 07:00 and 18:00. The time range in which energy consumption is primarily concentrated will be referred to as “working hours” from now on.



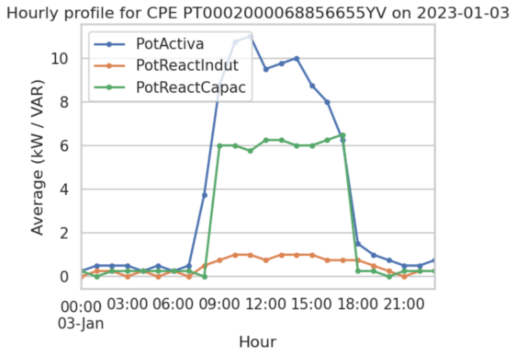


Figura 10: Profiles of the average daily energy for a single CPE within a month (January 2023) and the hourly average values for one day (March 1, 2023).

In the same way, starting from a selected CPE, additional profiles have been analyzed (Figure 11) to determine whether useful insights can be extracted from their patterns:

- Average Monthly Active Power for selected CPE
- Average Daily Active Power per Month for selected CPE
- Average Hourly Active Power per Month for selected CPE
- Weekday vs Weekend Average Consumption for selected CPE

These profiles reveal that energy consumption may depend on seasonal variations and that differences between weekdays and weekends can provide valuable information.

The following features have been selected from the above-mentioned considerations: **average peak value and time**, **peak to average ratio**, **working period**, **seasonal consumption ratio**, **weekday on weekend ratio** and **nighttime/worktime load share** as the fraction of consumption during night/worktime over total day.

Maia Municipality site Starting from a graph describing the charging of electric vehicles at the Maia Municipality site (`https://baze.cm-maia.pt/BaZe/HGP.htm`), it becomes evident that different CPEs exhibit distinct behaviors during daytime hours. Some CPEs operate within a relatively fixed working-hour range (e.g., around eight consecutive hours), while others display a more periodic consumption pattern, characterized by multiple shorter charging sessions distributed throughout the day. **Standard deviation** over the average day can define the CPE behavior. Other two good parameters that can help to define the distribution over the daily hours are **number of hours with energy value higher than 75% of peak average value**, **number of hours with energy value lower than 25%** and than **50%**. **Average value below/above threshold ratio** (25%) can also help to understand better the working hours behavior.

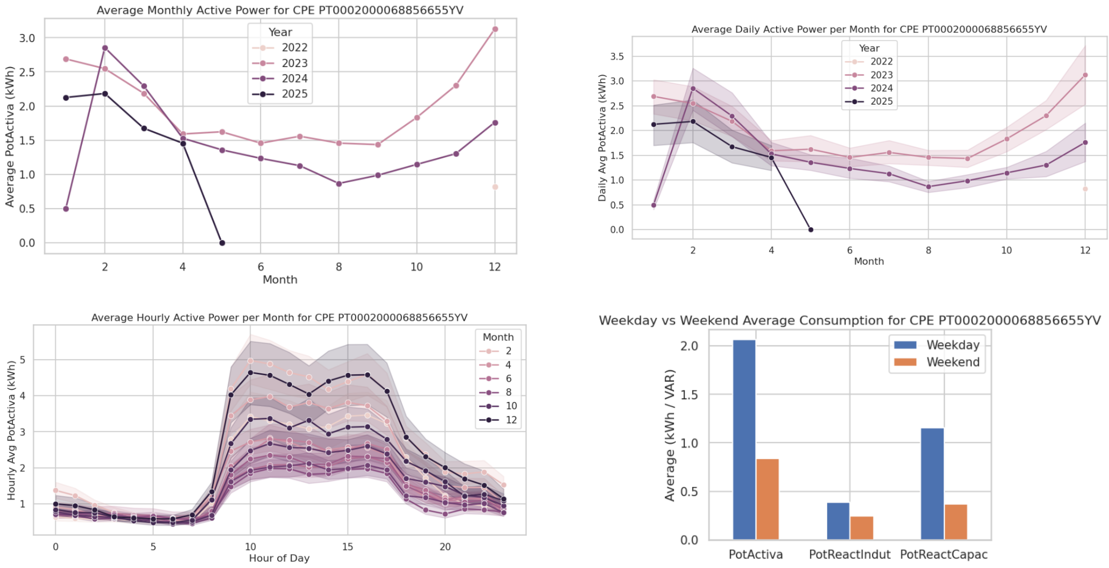


Figura 11: Profile average values for a single CPE

Correlation parameters Starting from figure 12, it is evident how is necessary to introduce a parameter to describe correlation between the different energy contributions.

Power factor $r = \frac{P}{\sqrt{P^2+(I+C)^2}}$ plays this role correlating Active power (P), Reactive Inductance (I) and Capacity (C). Specific correlation between Active power and the two singular contributions can be taken into account by respectively **power over inductance** $i = \frac{I}{\sqrt{P^2+(C+I)^2}}$ and **power over capacity** $c = \frac{C}{\sqrt{P^2+(C+I)^2}}$.

Correlation matrix A correlation matrix (Figure 13) was used to investigate the proposed features and assess whether each of them provides meaningful information. Figure 14 helps in correlation study.

The standard deviation (load_variability_index) is strongly correlated with the peak-to-average

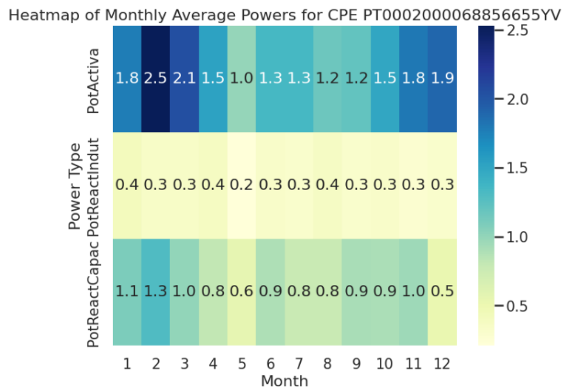


Figura 12: Heatmap of Monthly Average Powers for one CPE

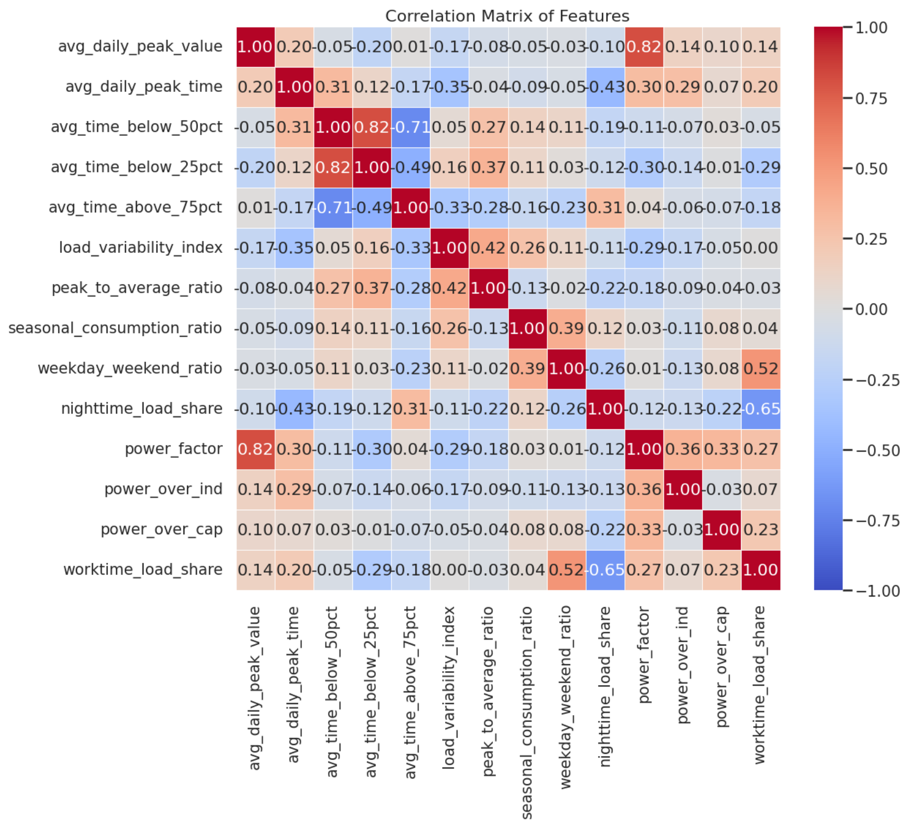


Figura 13: Features correlation matrix

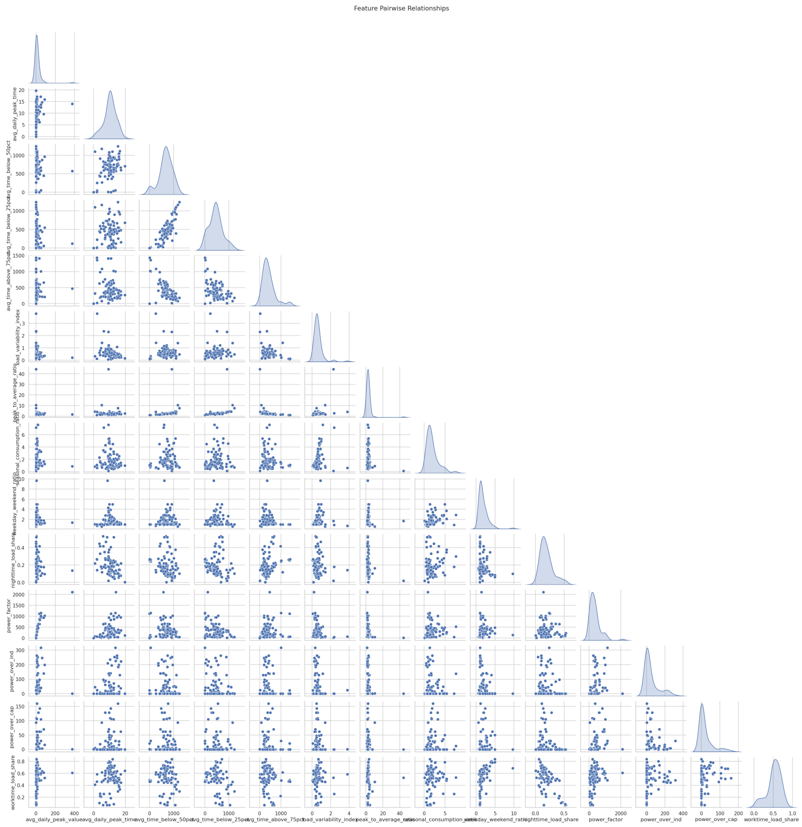


Figura 14: Feature pairwise relationships

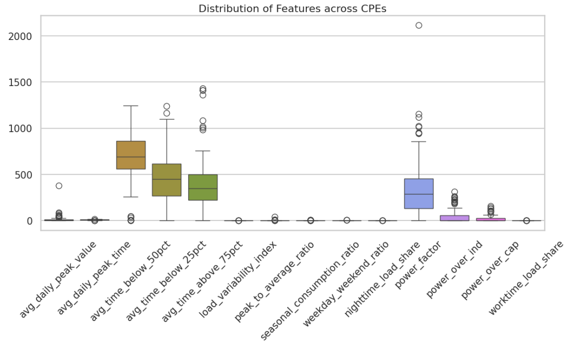


Figura 15: Feautures boxplot for all the CPE

ratio (peak_to_average_ratio), as both metrics describe deviations from the mean value. Conversely, the number of hours above 75% of the average peak (avg_time_above_75pct) appears to be a useful feature, given its low correlation with other parameters. On the other hand, the number of hours below 50% (avg_time_below_50pct) and below 25% (avg_time_below_25pct) are highly correlated, indicating that most data points fall below the 25% threshold. This also explains why avg_time_below_25pct is more correlated with peak_to_average_ratio than avg_time_below_50pct: values below 25% dominate the dataset and therefore have a stronger impact on the average. Based on these observations, we will retain only avg_time_above_75pct and avg_time_below_25pct for further analysis.
The parameters power_over_ind and power_over_cap show correlation with both power_factor and the average daily peak time. The correlation with power_factor is similar for the two contributions, indicating that they influence the total active power in a comparable way. Furthermore, the correlation between power_over_cap and workload_time_period could provide useful insights during the prediction phase.
Finally, power factor and average peak value are strongly correlated, which means the amount of info they bring are similar. Since the correlation between power factor and the other parameters is higher, we'll keep this parameter discarding avg_daily_peak_value.

Boxplot of the feautures Figure 15 summarizes the distribution of feature values across CPEs. Outliers are clearly visible, and the variability and asymmetry of the features can be observed. The high medians and wide dispersion of the parameters avg_time_below_50pct and avg_time_below_25pct confirm that CPEs spend the majority of time below 25% of the average peak value. Conversely, high load peaks (avg_time_above_75pct) are much less frequent. The parameters power_over_ind and power_over_cap exhibit low median values with a few outliers, indicating that only a small number of CPEs have a significant reactive contribution. This information can be useful for identifying peculiar cases.

Time window definition Four time windows have been defined to help the next clustering steps:

- Night: 00h - 06.00h
- Morning: 06.00h - 12.00h
- Afternoon: 12.00h - 18.00h
- Evening: 18.00h - 24.00h

TimeWindow	Afternoon_12_18	Evening_18_24	Morning_06_12	Night_00_06
CPE				
PT0002000068856655YV	2.964538	1.386926	1.821590	0.691944
PT0002000068856781NM	33.534233	30.841255	31.870941	30.561277
PT0002000068856872QG	0.149746	0.209996	0.113214	0.088811
PT0002000068856906VS	6.470755	5.709116	5.410211	5.732174
PT0002000068857099AR	2.237774	6.214691	1.863214	1.682610

Figura 16: Time windows definition for each CPE

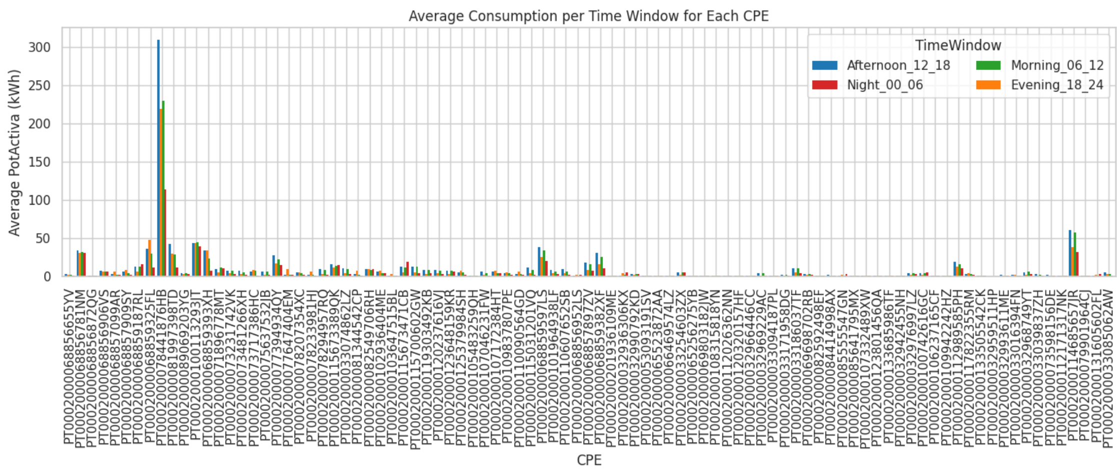


Figura 17: Average Consumption per Time Window for each CPE

and active power have been calculated for each CPE for each time window, in order to define some total energy trend in the different windows. Figure 16 represents this division for the first five CPEs.

Figure 17 summarizes the active power for each time window for each CPE. For each of these time window a boxplot can depict power consumption across CPE (figure 18). Removing the outliers can help to see the trend of the average power in the different time windows, as in figure 19, where is possible to see a lower overall energy consumption during the night and in the evening.

3 Modeling and evaluation

4 Exercise 1 - clustering

Clustering phase now will be performed by using k-means and DBSCAN methods.

K-means As first step, inf, $-inf$ and $NULL$ value has to be managed since k-means method does not support this values. In particular, $NULL$ values often appears in the dataset. They are substituted by 0.

K-Means is sensitive to outliers, which can significantly affect cluster centroids. Outlier detection has been performed by IQR method with IQR factor of 3.0 and they are removed from the dataset.

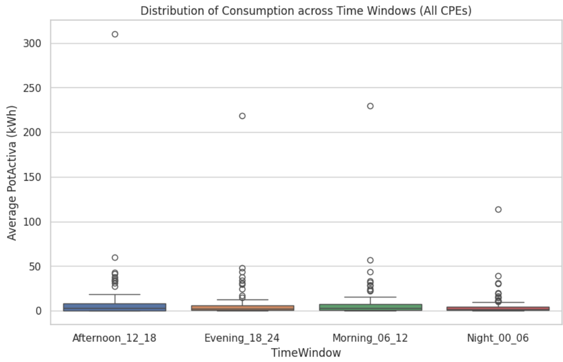


Figura 18: Distribution of power consumption across time windows

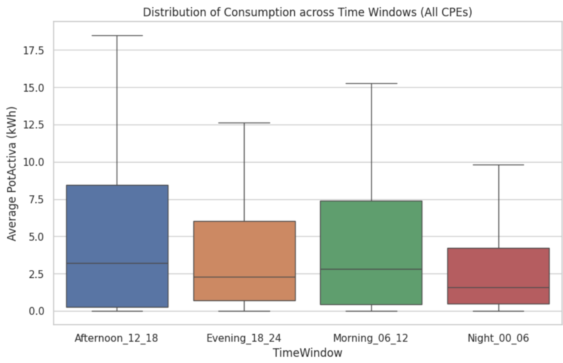


Figura 19: Distribution of power consumption across time windows without outliers

23 outliers are identified. Outliers will later be assigned to the nearest cluster centroid.

Different values of k in the range [2–15] were evaluated using the silhouette score as primary performance metric. The best value was found to be k = 11, and Figure 20 illustrates the distribution of CPEs across clusters.

A quality check was then performed on the K-Means results. For each cluster, the following metrics were computed: silhouette score, cluster size, standard deviation, minimum and maximum distance to the centroid, and the 25th, 50th, and 75th percentiles of these distances (Figure 21). The presence of very small clusters (e.g., cluster 6), clusters with negative silhouette values, and clusters with high mean distances (e.g., cluster 10) suggests that some clusters should be merged or split.

```
cluster_kmeans
1.0     17
0.0     14
4.0     13
3.0      9
7.0      8
10.0     8
2.0      8
6.0      6
9.0      4
8.0      3
5.0      1
```

Figura 20: $k = 11$, clusters definition.

```
Mean silhouette per cluster:
cluster
0     0.275
1    -0.022
2     0.472
3     0.215
4     0.402
5     0.410
6     0.668
7     0.215
8     0.225
9     0.233
10    0.053
Name: silhouette, dtype: float64

Distance-to-centroid distribution per cluster:
         count   mean    std    min    25%    50%    75%    max
cluster
0         10.0  1.504  0.419  0.888  1.277  1.427  1.785  2.176
1          6.0  2.860  1.326  1.575  1.890  2.388  4.001  4.545
2          7.0  1.468  0.412  0.827  1.260  1.576  1.636  2.082
3         13.0  1.741  0.533  0.996  1.400  1.775  1.961  3.038
4          3.0  1.762  0.198  1.621  1.649  1.678  1.833  1.988
5          5.0  1.507  0.367  0.881  1.502  1.622  1.751  1.781
6          2.0  0.641  0.000  0.641  0.641  0.641  0.641  0.641
7          4.0  1.645  0.680  0.988  1.140  1.567  2.072  2.457
8          6.0  1.808  0.545  0.972  1.483  1.942  2.169  2.415
9          7.0  1.692  0.624  0.857  1.269  1.604  2.123  2.599
10         5.0  2.281  0.872  1.040  2.230  2.249  2.385  3.500
```

Figura 21: quality checks and diagnostics

Clusters were flagged for merging or splitting based on the following guidelines:

- If a cluster is not sufficiently compact, it is considered poorly separated relative to its spread.
- If a cluster is too small, it is not reliable for characterization. In this case, the nearest centroid is used as a merge candidate.

To improve segmentation, the range of k was re-evaluated using multiple criteria: high silhouette score, low Davies–Bouldin index, high Calinski–Harabasz score, and the presence of a clear elbow in the inertia curve.
By applying these steps and enforcing a minimum cluster size of 3 elements, the new optimal value for k was 8. Increasing the minimum size to 4 elements resulted in k = 3, which was considered too small to provide meaningful segmentation. Figure 22 shows resulting plot by usign PCA method for dimensionality reduction.

Interpretation of formed clusters As first steps, the profiling of centroids is performed through z-score. High z-score means feature significantly above average, while low z-score means feature significantly below average.

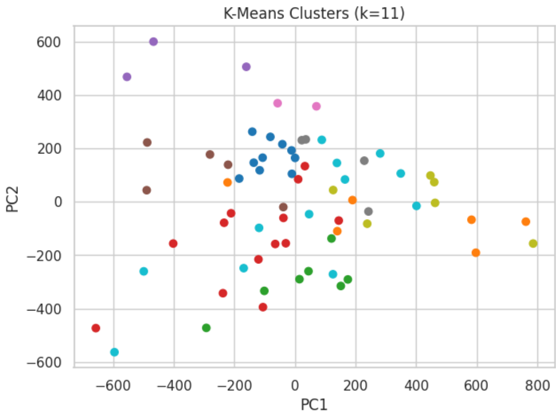


Figura 22: k-means representation

```
Cluster 0: top features -> ['worktime_load_share', 'weekday_weekend_ratio', 'power_factor', 'nighttime_load_share', 'power_over_ind']
Cluster 1: top features -> ['avg_daily_peak_time', 'worktime_load_share', 'weekday_weekend_ratio', 'power_factor', 'avg_time_above_75pct']
Cluster 2: top features -> ['nighttime_load_share', 'worktime_load_share', 'avg_time_above_75pct', 'weekday_weekend_ratio', 'avg_daily_peak_time']
Cluster 3: top features -> ['avg_daily_peak_value', 'power_factor', 'power_over_ind', 'power_over_cap', 'avg_time_below_25pct']
Cluster 4: top features -> ['avg_daily_peak_value', 'power_factor', 'power_over_cap', 'avg_time_below_25pct', 'avg_daily_peak_time']
Cluster 5: top features -> ['power_over_ind', 'avg_time_below_25pct', 'avg_daily_peak_time', 'seasonal_consumption_ratio', 'load_variability_index']
Cluster 6: top features -> ['avg_daily_peak_value', 'power_factor', 'nighttime_load_share', 'seasonal_consumption_ratio', 'avg_daily_peak_time']
Cluster 7: top features -> ['weekday_weekend_ratio', 'worktime_load_share', 'power_over_cap', 'nighttime_load_share', 'load_variability_index']
Cluster 8: top features -> ['avg_time_below_25pct', 'avg_daily_peak_time', 'nighttime_load_share', 'load_variability_index', 'weekday_weekend_ratio']
Cluster 9: top features -> ['seasonal_consumption_ratio', 'weekday_weekend_ratio', 'load_variability_index', 'avg_time_above_75pct', 'avg_time_below_25pct']
Cluster 10: top features -> ['load_variability_index', 'avg_time_above_75pct', 'power_factor', 'avg_daily_peak_value', 'avg_time_below_25pct']
```

Figura 23: top 5 feutures list for each cluster

Then, key features are identified for each cluster for each cluster by keeping top 5 z-scores features. Figure 23 enlists the top features for each cluster, while figure 24 gives an overall view with an heatmap of features for each cluster and z-score.

DBSCAN

Data preparation As first step, inf and $-inf$ and $NULL$ value has to be managed since clustering algorithms such as DBSCAN do not support these values. In particular, $NULL$ values often appears in the dataset. They are substituted by 0.

Epsilon-selection To determine a suitable value for epsilon, we computed the distances to the k-th nearest neighbor (with k=3) for all points in the dataset after standardization. These distances were sorted and analyzed to identify the point of maximum curvature (the "knee") in the k-distance plot. This knee represents a transition between dense regions and sparser areas, and is commonly used as a reference for setting epsilon. We then explored a range of epsilon values around and beyond this knee to evaluate clustering behavior and select the configuration that maximizes the number of clusters. The resulting clusters are shown in Figure 25..

Overall features of clusters can be seen in 26, where colors of the heatmap are defined by the effect size E_c, f for the cluster c and the feature f:

$$E_{c,f} = \frac{\mathrm{median}\big(X_f \mid \mathrm{cluster} = c\big) \;-\; \mathrm{median}\big(X_f\big)}{\mathrm{IQR}\big(X_f\big)} \tag{1}$$

which is the difference between the median value of feature f considering only the points in cluster c and the global median of feature f across all data points, normalized by the global Interquartile Range (IQR) of feature f, which represents the spread of the middle 50% of its

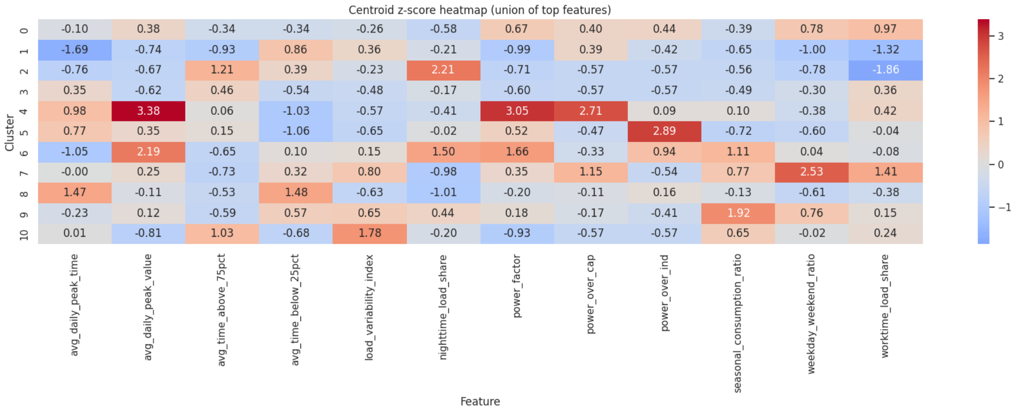


Figura 24: k-means heat map (based on z-score)

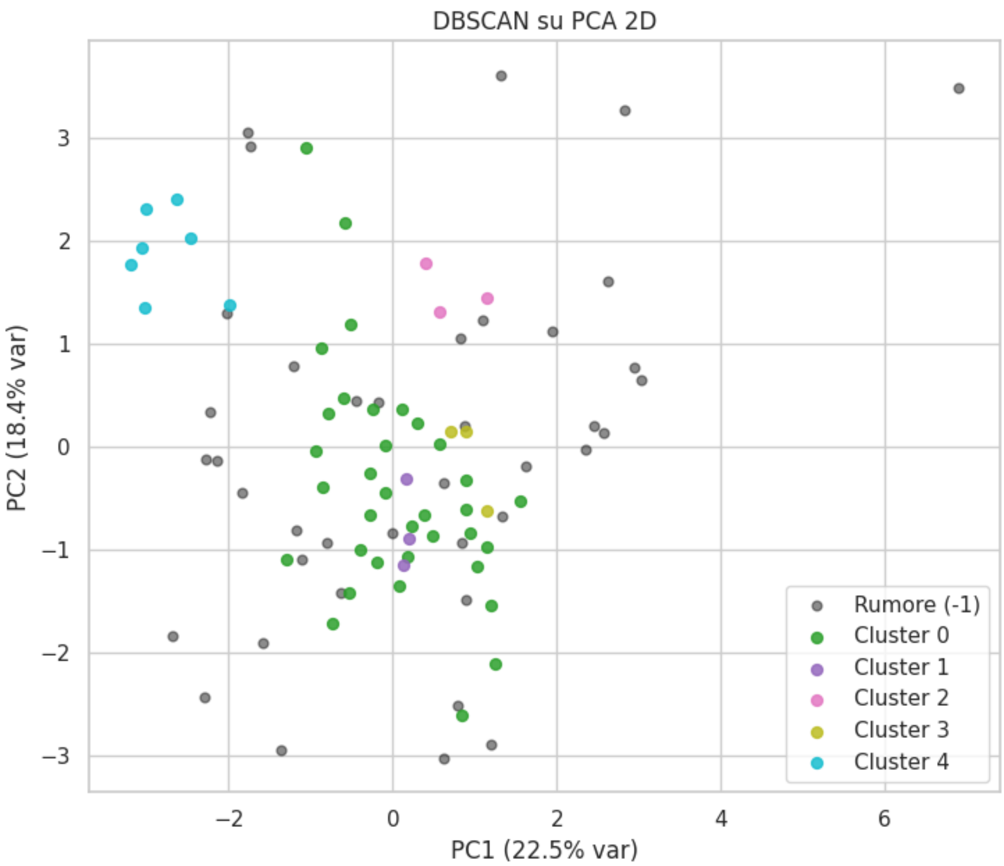


Figura 25: DBSCAN clustering with $\epsilon = 1.7684$ and a minimum core points of 3.

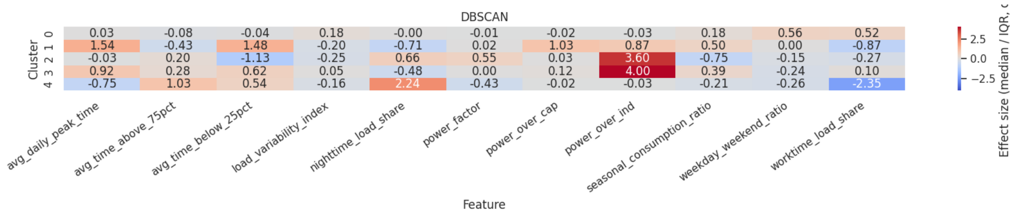


Figura 26: DBSCAN clusters heat map (based on effect size)

values.
If the effect size is greater than 0, cluster c has a higher median for feature f than the global median. If it is lower, cluster c has a lower median for feature f.

5 Exercise 2 - supervised learning

The objective of Experiment 2 is to systematically evaluate supervised learning approaches for **one-week-ahead active power forecasting** at the individual CPE level. The experiment compares **pure time-series models** with **feature-based regression models**, assessing whether increased model complexity leads to meaningful improvements over a **simple weekly persistence baseline**. The analysis is conducted in a controlled and comparable setting, ensuring consistency across CPEs in terms of **data resolution, forecasting horizon, and evaluation metrics**.

Temporal Resolution and Signal Characteristics The original dataset provides energy measurements at a **15-minute resolution**. While this granularity captures fine-scale fluctuations, it also introduces substantial **high-frequency noise**, which can obscure the underlying consumption dynamics. This effect is clearly visible in figure 27 and 28, where both Random Forest and XGBoost models exhibit highly erratic predictions when trained on 15-minute data.

When the same series are aggregated to an **hourly resolution**, the signal becomes smoother and more structured. Hourly aggregation reduces short-term volatility, improves learning stability, and leads to lower normalized prediction errors. Moreover, the computational burden of training time-series models—particularly ARIMA—is significantly reduced, whereas applying ARIMA to 15-minute data becomes impractical due to excessive series length.

Figure 27 provides a direct numerical comparison between 15-minute and hourly resolutions for the XGBoost model over the same one-week test window. At 15-minute resolution, the weekly baseline achieves an NRMSE of **0.925**, while XGBoost reduces the error to **0.653**, indicating that machine learning can partially mitigate high-frequency noise but still operates under considerable variability.

After hourly aggregation, the baseline error drops to **0.482**, corresponding to a **48% relative reduction** compared to the 15-minute baseline. In contrast, hourly XGBoost reaches an NRMSE of **0.525**, slightly worse than the baseline. This result confirms that once high-frequency noise is removed, the dominant weekly structure is already well captured by the persistence rule, leaving limited room for improvement through increased model complexity.

Overall, **figure 27 quantitatively demonstrates that temporal aggregation contributes more to error reduction than model complexity**, particularly in the presence of strong weekly seasonality.

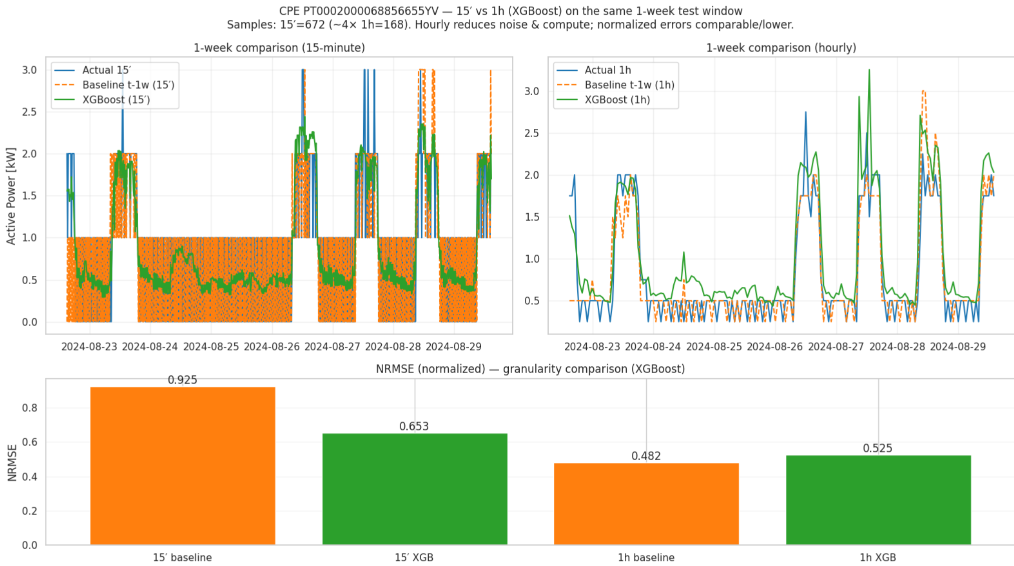


Figura 27: Comparison between 15 minutes and hourly based data: XGBoost approach

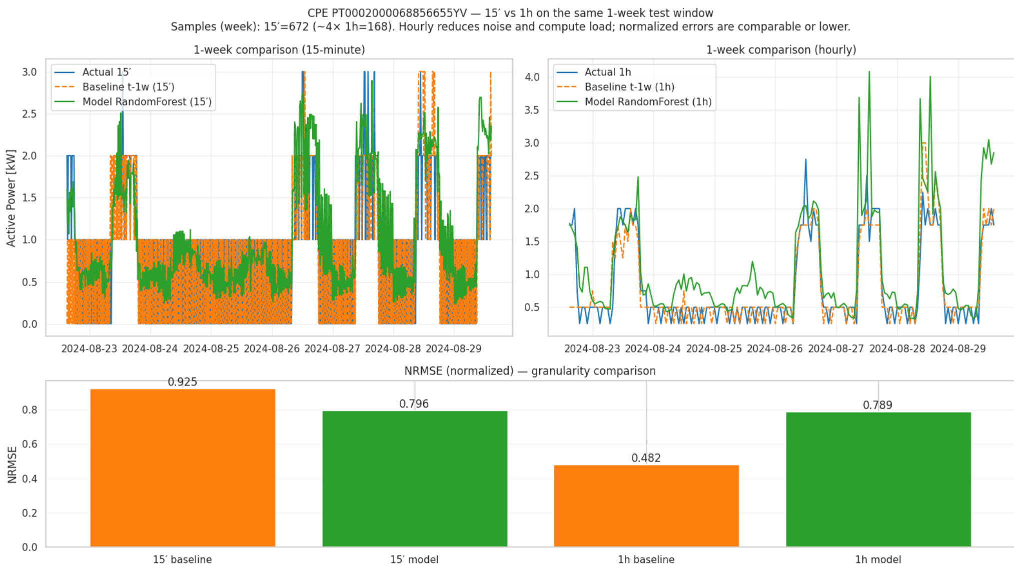


Figura 28: Comparison between 15 minutes and hourly based data: RandomForest approach

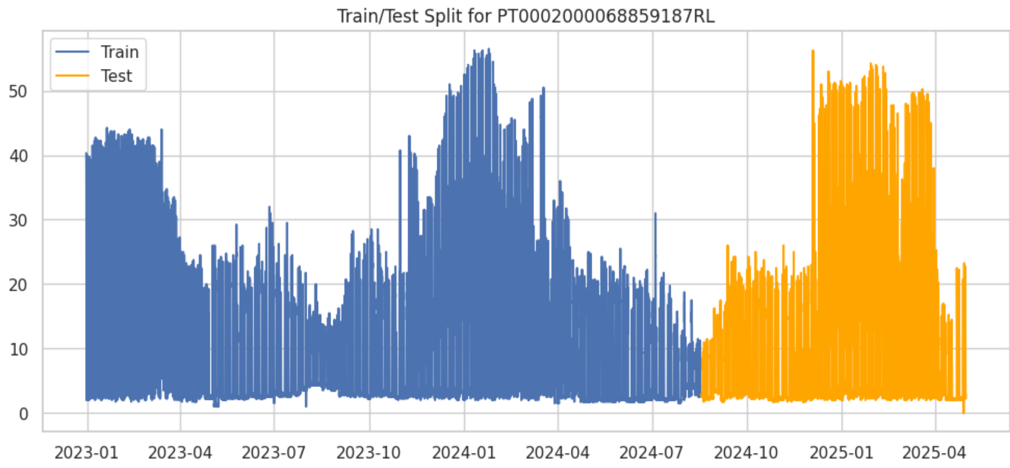


Figura 29: Temporal separation for time-series model (one CPE as example)

5.1 Time-Series Modeling: ARIMA and Weekly Persistence Baseline

For each CPE, the hourly time series is sorted chronologically and split using a **temporal criterion**, with the first **70% of observations used for training** and the remaining **30% reserved for testing**, as illustrated in Figure 29. Model evaluation focuses on the **first week (168 hours)** of the test period to ensure consistency across CPEs. A **weekly persistence baseline** is defined as the consumption observed at the same hour exactly one week earlier. This baseline is particularly relevant in energy forecasting contexts, where consumption patterns are strongly influenced by weekly operational schedules. An ARIMA model is trained on the training portion of each CPE's time series and used to generate forecasts for the test window. Performance is assessed using error-based metrics, with a primary focus on **RMSE and normalized RMSE**. Figure 29 illustrates the chronological 70/30 train–test split applied to a representative CPE. The clear temporal separation ensures that all evaluated models operate under realistic forecasting conditions, with **no leakage of future information** into the training phase. The test period corresponds to a distinct temporal regime characterized by higher variability and pronounced peaks. Despite this shift, both the baseline and ARIMA models maintain stable performance, further supporting the conclusion that **weekly repetition remains the dominant signal**, even under non-stationary conditions. Across many CPEs, ARIMA does not consistently outperform the weekly baseline, suggesting that the dominant structure in the data is **strong weekly seasonality**, which is already effectively captured by the persistence rule. In such cases, the marginal gains from modeling additional autoregressive dynamics are limited.

5.2 Feature-Based Supervised Learning Framework

To address the limitations of univariate time-series models, a **feature-based supervised learning framework** is introduced. A hybrid dataset is constructed by combining:

- **Static features**, including DBSCAN cluster assignments and energy profile indicators;
- **Temporal features** derived from the hourly series, such as weekly lags at 168 and 336 hours and lagged rolling means;
- **Calendar features**, including hour-of-day and day-of-week indicators.

To prevent information leakage, all lagged features are constrained to delays of **at least 168 hours**, ensuring that only past information relative to the forecast horizon is used. The dataset is then split chronologically into **70% training and 30% testing data**. The following models are evaluated under identical experimental conditions: **Weekly Baseline, Random Forest, XGBoost, MLP** (with feature scaling via StandardScaler), and **ARIMA** trained solely on the target series.

5.3 Model Comparison and Cluster-Level Analysis

For each CPE, the model achieving the **lowest RMSE** on the test period is identified as the best-performing approach. These results are aggregated at the cluster level to investigate whether forecasting performance depends on **consumption behavior rather than model class alone**. The distribution of best-performing models across clusters is summarized in Figure 30.

The analysis reveals clear and consistent patterns. Clusters characterized by **regular and repetitive consumption profiles** are best predicted by the weekly persistence baseline, which effectively captures the dominant seasonal structure. In contrast, clusters exhibiting **irregular, heterogeneous, or weakly periodic behavior** benefit from feature-based machine learning models, particularly tree-based methods capable of modeling nonlinear relationships. Figure 30 quantitatively highlights strong heterogeneity across clusters:

- **Cluster 3 (strictly periodic profiles)**: the weekly baseline dominates with **71.4%** of CPEs.
- **Cluster 0 (regular but non-deterministic)**: the baseline is best for **66.7%**, with Random Forest covering the remaining **33.3%**.
- **Cluster 1 (highly irregular)**: XGBoost dominates with **51.4%**, while the baseline drops to **18.9%**.
- **Cluster 1 (noise/outliers)**: no single model dominates; XGBoost leads with **44.7%**, followed by the baseline at **31.6%**.
- **Cluster 4 (semi-regular)**: performance is evenly split (**33.3%**) between ARIMA, baseline, and MLP.

These results confirm that **model effectiveness is strongly cluster-dependent**, and that no global best model exists across heterogeneous consumption patterns. In clusters with relatively simple but stable dynamics, **ARIMA remains competitive or superior** to feature-based approaches.

5.4 Interpretation of Results

The results of Experiment 2 indicate that forecasting performance is primarily driven by the statistical properties of the consumption profiles, rather than by model complexity alone. The strong performance of the weekly baseline reflects the presence of dominant weekly seasonality across a large portion of the dataset. From a bias–variance perspective, the baseline acts as a low-variance estimator, delivering stable predictions without overfitting noise.Feature-based machine learning models add value mainly when weekly regularity weakens and auxiliary features provide additional explanatory power. However, these gains are **cluster-specific** rather than universal.

5.5 Implications for Subsequent Experiments

Experiment 2 demonstrates that a **single global forecasting model** is suboptimal for heterogeneous energy consumers. The strong dependence of model performance on cluster membership motivates the cluster-aware normalization and model-selection strategies developed in Experiment 3. By tailoring preprocessing and modeling choices to the statistical characteristics of each cluster, more robust, interpretable, and efficient forecasting pipelines can be constructed.

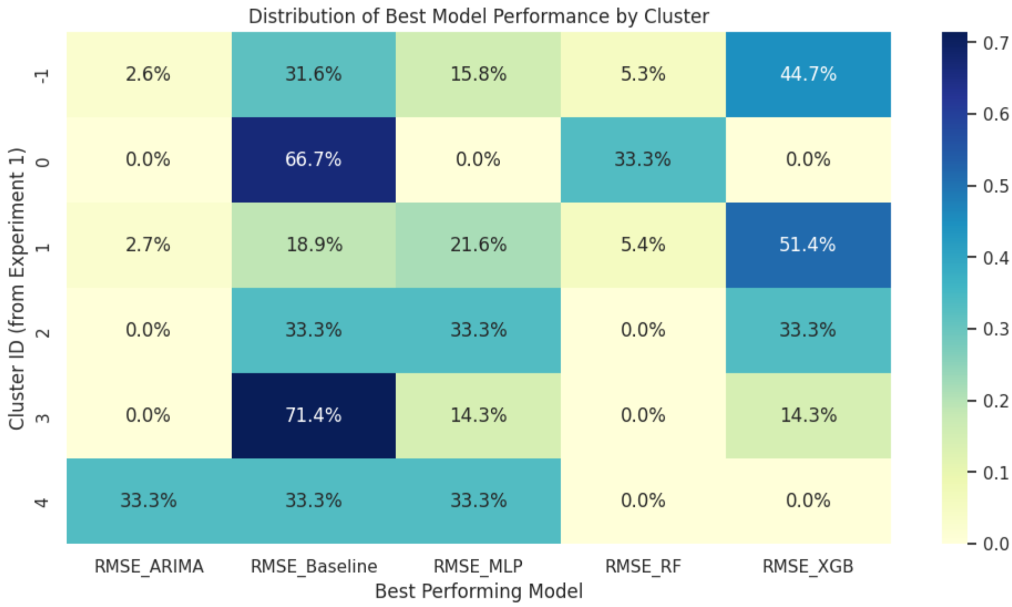


Figura 30: Heatmap of best model performance per cluster (DBSCAN clustering)

6 Experiment 3 - Comparative Evaluation by Normalization

The primary objective of Experiment 3 was to refine the forecasting accuracy by tailoring regression models to the specific consumption patterns identified in the clustering phase (Exercise 1). Instead of applying a *one-size-fits-all* model, this experiment sought to determine which specific combination of machine learning algorithm and data normalization technique yields the lowest Root Mean Squared Error (RMSE) for each customer cluster.

Experimental Setup The experiment utilized a nested iterative approach to exhaustively test combinations of data segments, preprocessing steps, and model architectures. The procedure we used the following feature scaling and target variable transformations:

- **Input Scalers (S_X):** To assess the sensitivity of models to feature magnitude and outliers, four strategies were tested:
  - *None*: Raw feature usage.
  - *StandardScaler*: Standardization ($\mu = 0,\ \sigma = 1$).
  - *MinMaxScaler*: Normalization to a bounded interval $[0, 1]$.
  - *RobustScaler*: Scaling based on percentiles (robust to outliers).
- **Target Transformations (T_y):** To address heteroscedasticity and stationarity, the target variable y (Active Power) was subjected to four transformations prior to training:
  - *None*: Predicting raw values.
  - *Log1p*: Logarithmic transformation ($\log(1 + y)$) to compress the range of high-variance data.
  - *StandardScaler*: Standardizing the target distribution (fit on training targets).
  - *Seasonal*: Subtraction of the hourly-weekly profile to model residuals rather than absolute load.

6.1 Pipeline Evaluation

In this experiment, we extended the supervised learning framework to optimize the preprocessing and modeling pipeline for each specific cluster. The goal was to determine the optimal combination of **Input Scalers** (S_X), **Target Transformations** (T_y), and **Regression Models** (M) that minimizes the Root Mean Squared Error (RMSE).

Methodology We performed an exhaustive grid search starting for the target CPE and later we etend the search for all the CPEs, devided by cluster group.

Figure 31 presents the average RMSE, grouped by Model and Normalization method.

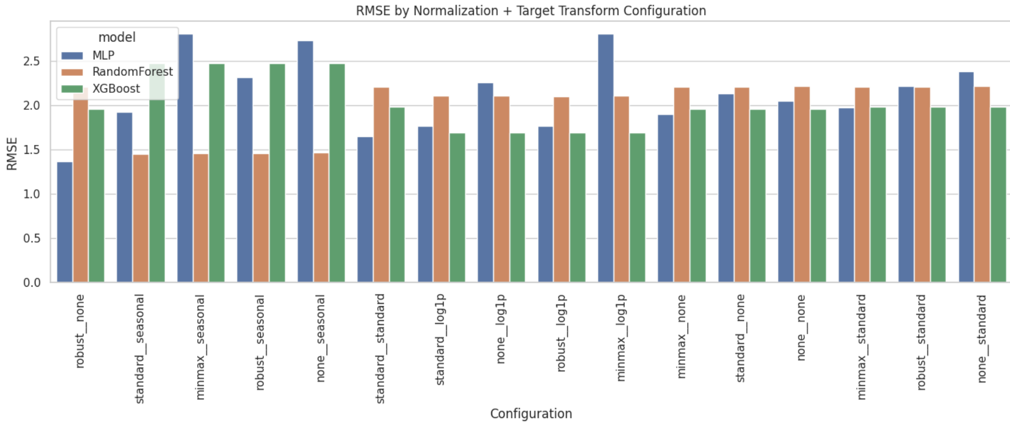


Figura 31: Average RMSE for target CPE, grouped by Model and Input Scaler. Tree-based models show higher stability across different scaling methods.

Commentary on Normalization Sensitivity: As observed in Figure 31, tree-based ensembles (Random Forest, XGBoost) demonstrate remarkable robustness to input scaling. The variance in RMSE between the *None*, *Standard*, and *MinMax* scalers is negligible for these models. In contrast, linear models and distance-based algorithms exhibit significant performance degradation when data is unscaled or improperly scaled, confirming the necessity of robust normalization (e.g., *RobustScaler*) when using non-tree architectures on data with outliers.

Figure 32 details the specific configuration that achieved the lowest error for each cluster.

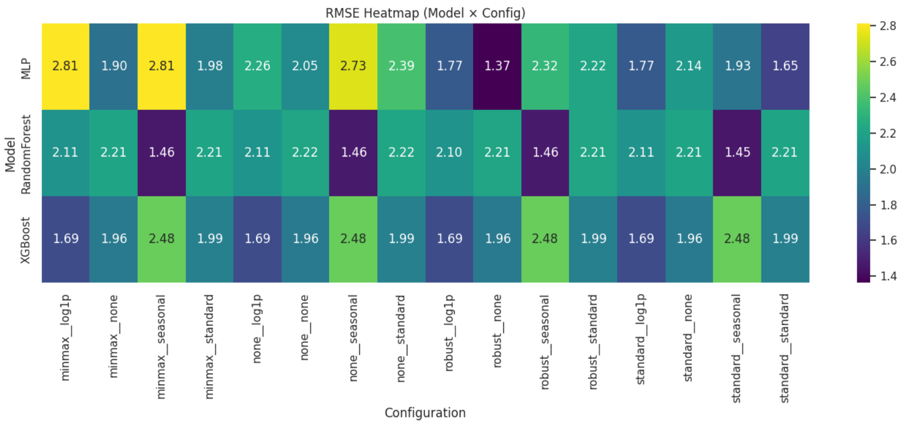


Figura 32: Optimal Configuration (Scaler + Target Transform + Model) selected for each cluster based on minimum RMSE.

Commentary on Target Transformations: The breakdown in Figure 32 highlights the impact of Target Transformations (T_y).

- The **Log1p** transformation frequently appears in the optimal configurations for clusters with high variance (i.e., "spiky"consumption), as it compresses the output space and prevents the model from over-penalizing peak errors.

- The **Seasonal** subtraction proves effective for clusters with highly regular commercial schedules, allowing the model to focus purely on predicting the residuals (deviations from the norm) rather than the raw load.

Conclusion: The results confirm that there is no single global best model. While XGBoost is often the strongest performer, the optimal preprocessing pipeline varies significantly by customer segment. This indicates that there is a need of a different approach.

6.2 Advanced Optimization

To further refine the forecasting accuracy, Experiment 3 expanded the search space for the target CPE, but this time, using per-timestamp features and not per-CPE aggregates. Following the previous steps we obtain Figure 33 which illustrates the Average RMSE across all clusters, grouped by Model and Input Scaler.

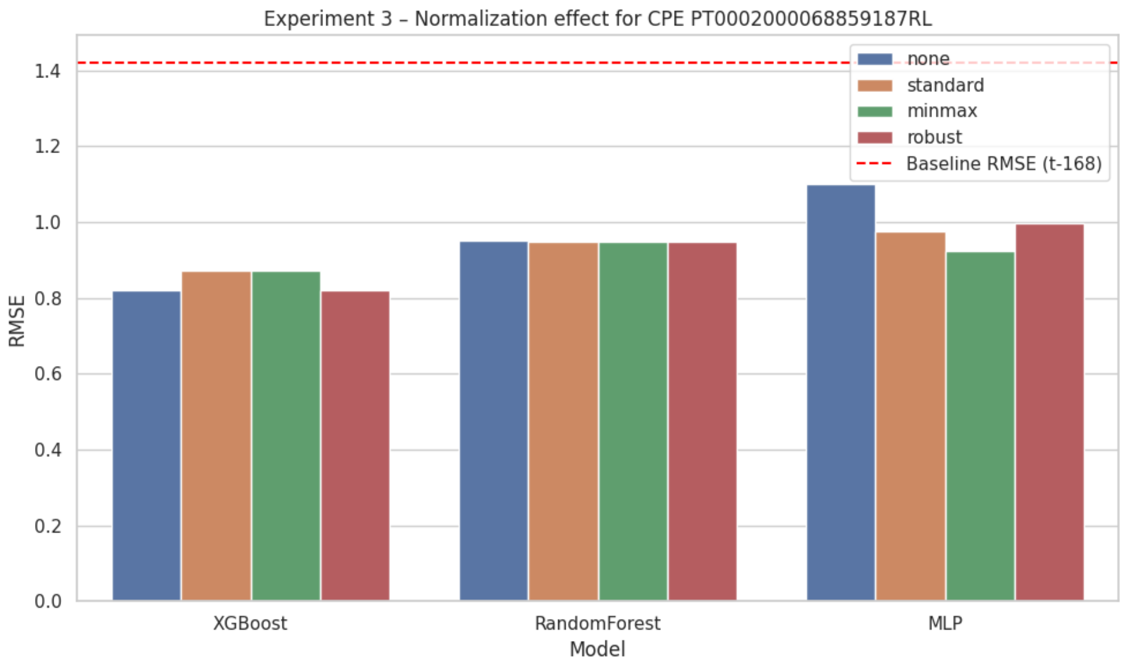


Figura 33: Impact of Input Scaling on Model Performance. Tree-based models (Random Forest, XGBoost) exhibit stability across all scalers, whereas Linear and Distance-based models show high sensitivity to data magnitude.

As depicted in Figure 33, tree-based ensemble methods (Random Forest, XGBoost) demonstrate significant robustness, maintaining low RMSE scores regardless of the scaling method used. This is expected, as decision trees split based on thresholds rather than Euclidean distances. In contrast, linear models (Linear Regression, Ridge, Lasso) and distance-based algorithms (KNN) suffer severe performance degradation when features are unscaled or when outliers are not managed (e.g., using *None* or *StandardScaler* on skewed data). The *RobustScaler* proved most effective for these sensitive models.

6.3 Cluster-Based Normalization and Modeling

Following the segmentation of consumers in Exercise 1, we adopted a strategy of **Cluster-Based Normalization**. Instead of training a single global model for the entire dataset, we treated each cluster as an independent dataset for the following reasons:

1. Mitigation of Statistical Heterogeneity The dataset contains varying types of consumers (e.g., large industrial facilities vs. small residential units) with varying orders of magnitude in power consumption.

- **Global Normalization Risk:** Applying a single scaler (e.g., MinMaxScaler) to the entire dataset would result in the "squashing"of data for smaller consumers. If a cluster with a peak of 5 kWh is normalized alongside a cluster with a peak of 500 kWh, the signals from the smaller cluster would become close to zero, effectively acting as noise to the model.

- **Local Normalization Solution:** By normalizing within each cluster, we preserve the relative shape and variance of the load curves for every consumer type, ensuring the model can learn patterns regardless of the absolute magnitude of consumption.

2. Specialization of Consumption Dynamics Different clusters exhibit conflicting temporal patterns. For example, "Commercial"clusters may show high consumption during the day and near-zero at night, while "Residential"clusters might peak in the evening. A global model would attempt to average these conflicting behaviors, likely resulting in mediocre performance for both. By segmenting the analysis, the regression algorithms (and the specific preprocessing steps like Seasonal Subtraction) can specialize in the specific seasonality and variance profile of that group.

3. Targeted Outlier Management As observed in the clustering phase, some groups are characterized by high variance and frequent outliers ("spiky"behavior), while others are stable. This segmentation allows us to apply aggressive transformations (like *Log1p* or *RobustScaler*) only to the clusters that require them, without distorting the data of stable clusters where such heavy-handed preprocessing is unnecessary.

The effectiveness of applying normalization strategies independently within each cluster is visually summarized in Figure 31. The results lead to two key observations. Firstly, the Baseline is still the dominant forecasting model without any further improvement after normalization. Secondly, after normalization we observe that Cluster 1 functions better by using the scaler minmax.

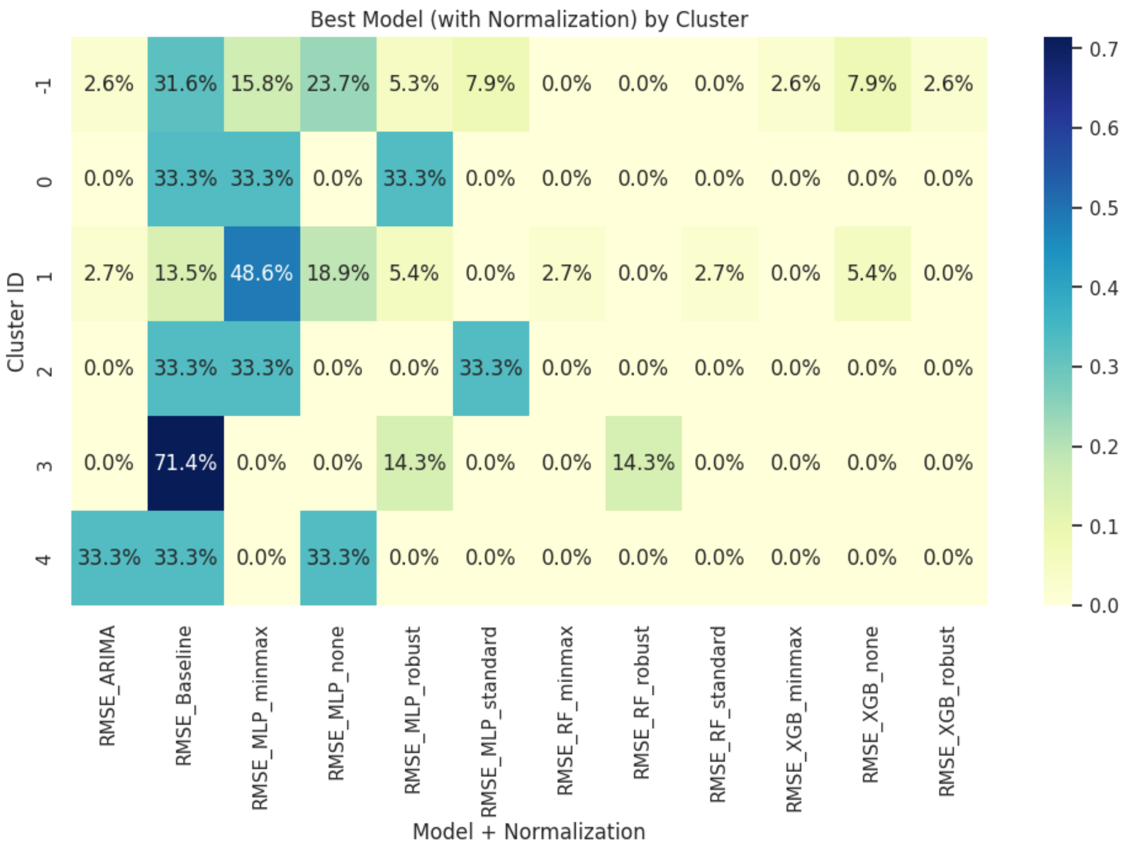


Figura 34: Comparison of Average RMSE by Model and Scaler.

Having a high accuracy level using the Baseline mode, one can suppose that this cluster referes to consumers with strong weekly repetition. We observe that cluster 3 uses as a best model the Baseline model, but can be used from -1, 0, 2 and 4 as well, despite that these clusters can perform well with different models. On the other half, cluster 1 using the minmax model, which helps the MLP learn non-linear patterns, refers to consumers who show important non-linear patterns, have daily cycles that differ from week to week and represent buildings with complex schedules and irregular peaks.

Cluster 4 is the only one to perfrom well using the ARIMA model as simple time-series modeling worked better than feature-based ML. Finally, Robust Scaler did not yield improvements in any cluster, suggesting that outliers were not a dominant issue in this dataset.

By identifying the minimum RMSE configuration for each cluster, we established a specific model-selection strategy. This ensures that future predictions for a specific CPE are generated using the algorithm best suited to its cluster's statistical properties, rather than the global average best model.

6.4 Forecast

In this section, we compare the forecasting performance of **15-minute resolution** versus **Hourly aggregation** across representative CPEs from each cluster.

The bar charts presented below display the **Normalized RMSE (NRMSE)** for four distinct scenarios:

- **15' Base:** Naive weekly baseline (persistence) at 15-minute resolution.
- **15' Model:** Random Forest prediction at 15-minute resolution.
- **1h Base:** Naive weekly baseline at Hourly resolution.
- **1h Model:** Random Forest prediction at Hourly resolution.

Note: Lower NRMSE indicates better performance.

Cluster -1 (Noise / Outliers)

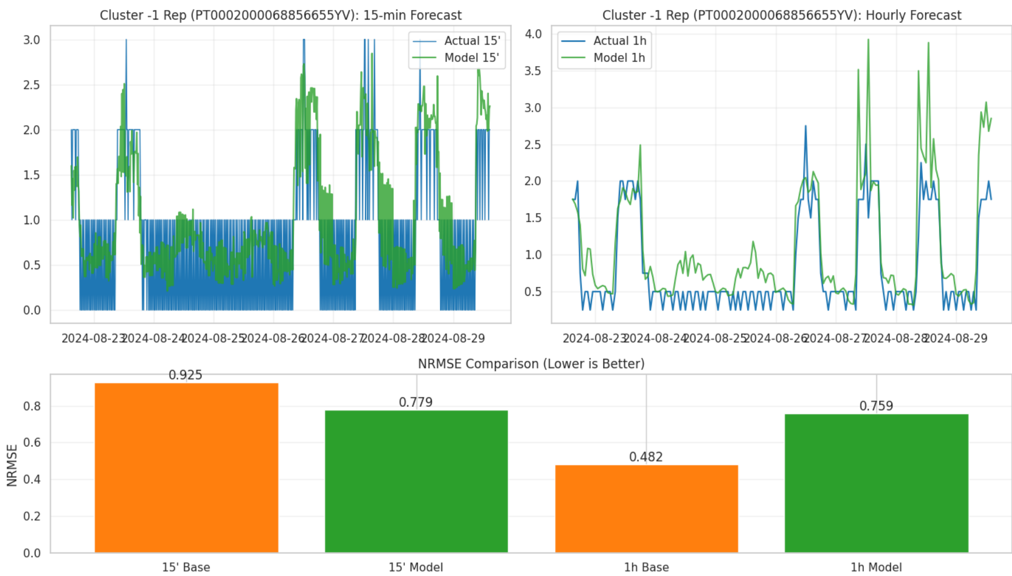


Figura 35: Forecasting results for Cluster -1 Representative

Observations:

- **Hourly Aggregation Win:** The transition from 15-minute to Hourly resolution reduces the baseline error significantly (0.925 → 0.482).
- **Baseline Superiority:** The naive baseline outperforms the ML model. The noisy nature of this cluster makes it difficult for the model to find a pattern better than "same time last week."

Cluster 0 (Regular Patterns)

Observations:

- **Clear Hourly Advantage:** The hourly baseline (0.377) is the best performing metric overall.
- **Model Struggle:** The model improves slightly when moving to hourly data (0.472 → 0.437) but still fails to beat the simple weekly lag.

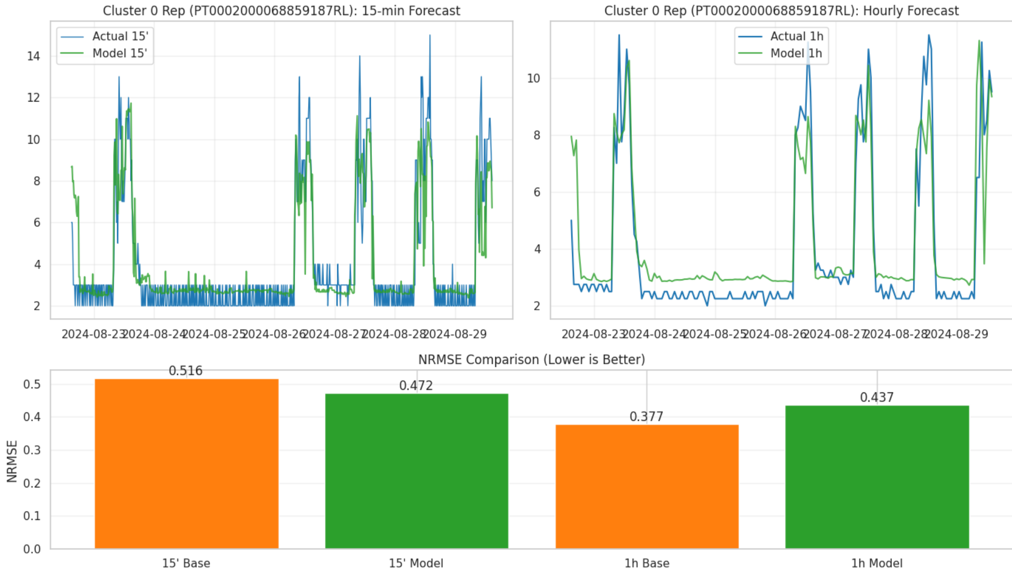


Figura 36: Forecasting results for Cluster 0 Representative

Cluster 1 (Highly Irregular)

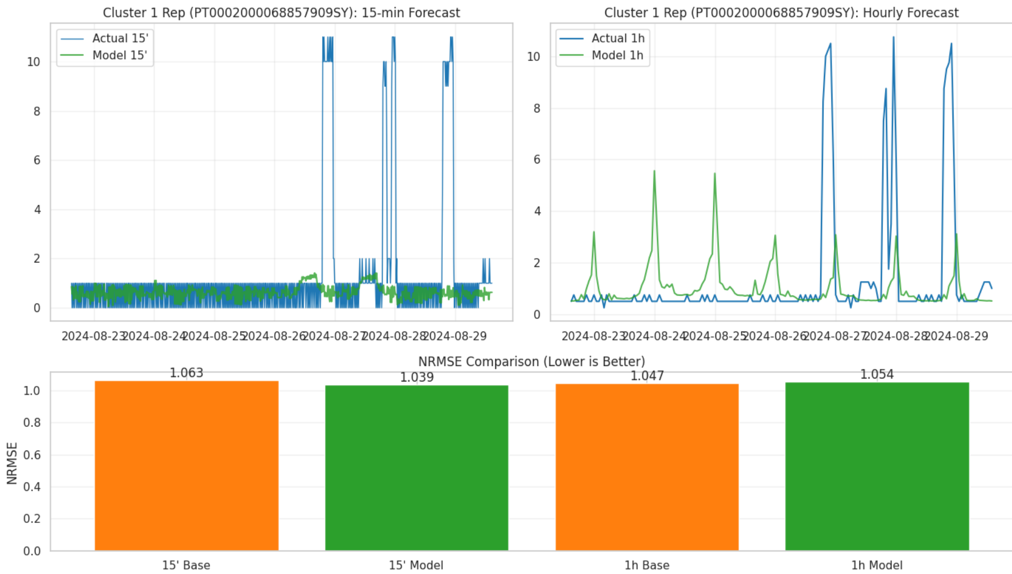


Figura 37: Forecasting results for Cluster 1 Representative

Observations:

- **High Error:** All NRMSE values are > 1.0, indicating that prediction for this cluster is extremely difficult (worse than predicting the mean).

- **No Aggregation Benefit:** Unlike other clusters, hourly aggregation does not help here. This suggests the volatility is inherent to the behavior (e.g., random spikes) rather than just high-frequency noise.

Cluster 2 (Volatile)

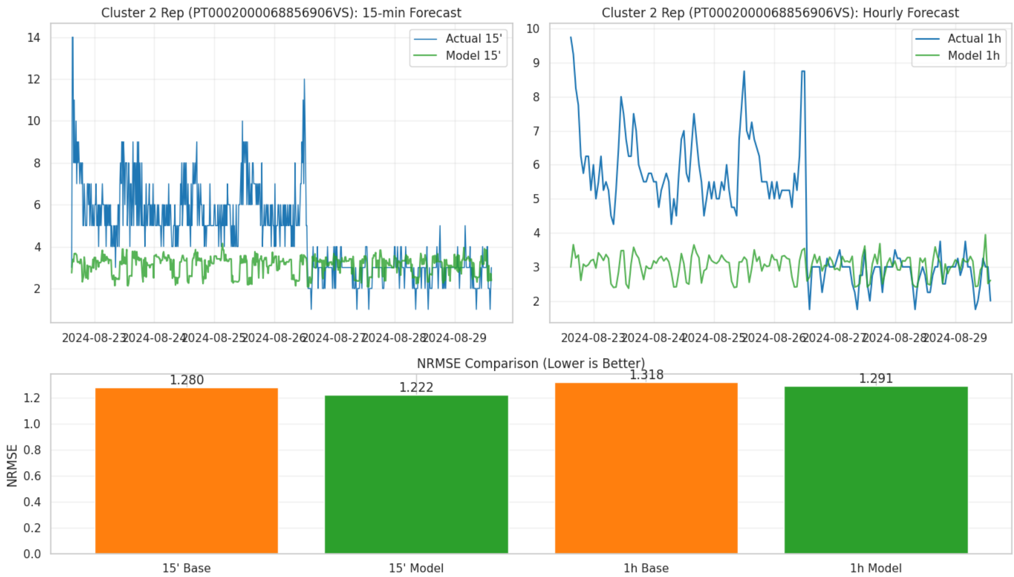


Figura 38: Forecasting results for Cluster 2 Representative

Observations:

- **Model Win:** This is one of the few cases where the **15' Model (1.222)** beats the 15' Baseline (1.280) and performs comparably to the hourly metrics.
- However, the overall error remains very high, similar to Cluster 1.

Cluster 3 (Strictly Periodic)

Observations:

- **The "Clockwork"Cluster:** The baseline error is almost zero (0.041), meaning this CPE repeats its behavior exactly week-over-week.
- **ML Failure:** The Machine Learning model performs terribly (NRMSE $>$ 0.9). This is a classic case where a complex model overfits or fails to capture a simple "on/off"step function, whereas a simple lag model is perfect.

Cluster 4 (Semi-Regular)

Observations:

- **Significant Noise Reduction:** Moving to hourly aggregation cuts the baseline error by more than half (0.920 $\rightarrow$ 0.420).

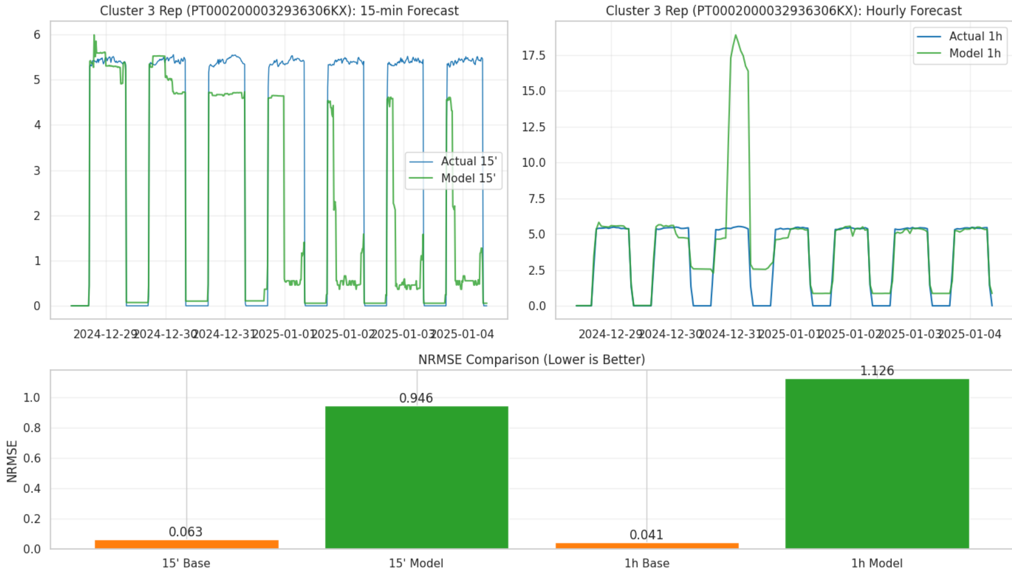


Figura 39: Forecasting results for Cluster 3 Representative

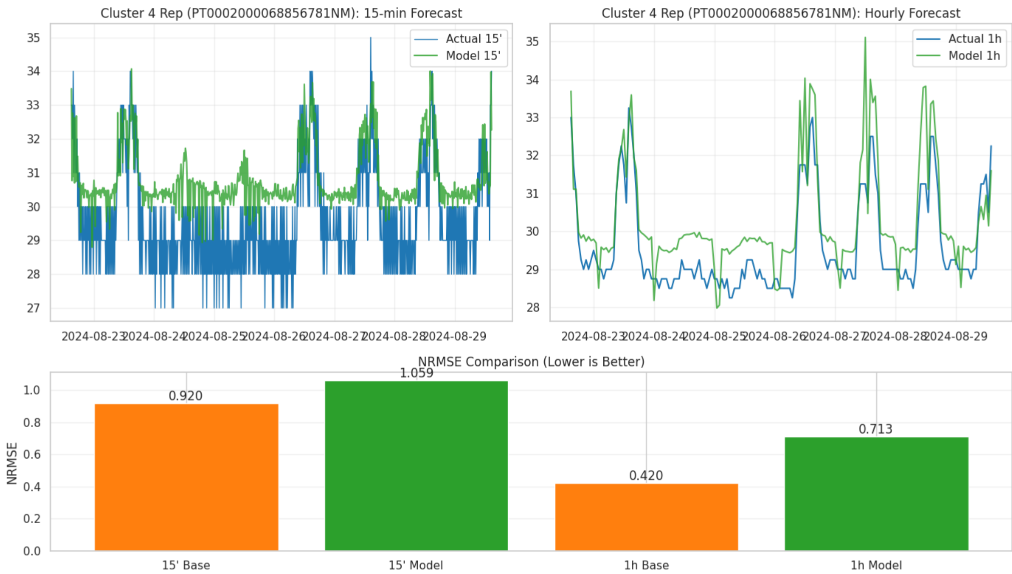


Figura 40: Forecasting results for Cluster 4 Representative

- **Recommendation:** For this cluster, the **Hourly Baseline** is the definitive winning strategy.

Summary of Findings

1. **Hourly > 15-Minute:** In almost all valid clusters (0, 4, -1), aggregating to 1-hour intervals drastically reduces the error (NRMSE), confirming that 15-minute data contains too much

noise for accurate forecasting.

2. **Baseline is the dominant model:** For this specific dataset, the **Weekly Persistence Baseline (Lag 168)** consistently outperforms the Random Forest model. This indicates the load profiles are dominated by strong weekly seasonality rather than complex non-linear interactions.

6.5 Analysis of K-Means Clustering Results

The application of K-means clustering ($k = 11$) revealed distinct performance patterns across different consumer profiles. This section analyzes the error metrics and examines specific cluster representatives to understand the forecasting behavior.

6.5.1 Normalized Error Analysis

Figure 41 presents the Normalized Root Mean Square Error (NRMSE) across all 11 clusters. Normalization allows for a fair comparison of predictability independent of the magnitude of energy consumption.

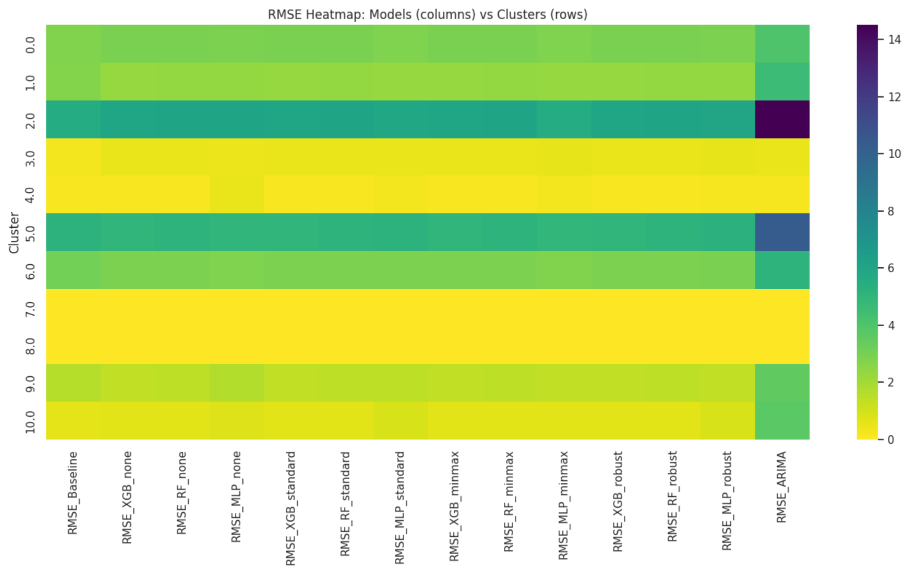


Figura 41: Normalized RMSE (NRMSE) across the 11 identified clusters.

Analysis: The NRMSE plot highlights a significant variance in predictability. Clusters with lower NRMSE values (e.g., Clusters 0 and 3) exhibit highly regular patterns that are easily captured by the Weekly Persistence Baseline. In contrast, specific clusters show NRMSE values exceeding 1.0, indicating that the variance of the error is comparable to or greater than the variance of the data itself. This suggests that for these specific subsets of CPEs, the consumption behavior is highly stochastic or driven by external factors not captured in the historical data.

6.5.2 Absolute Error Analysis

While NRMSE provides a view of relative performance, the absolute Root Mean Square Error (RMSE) shown in Figure 42 indicates the actual magnitude of the prediction errors in kWh.

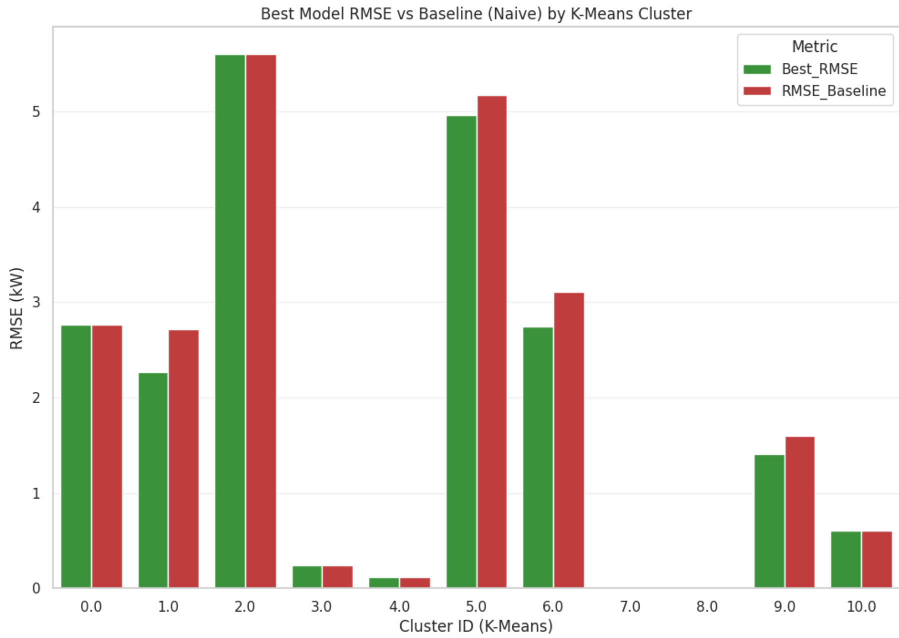


Figura 42: Absolute RMSE values across the 11 clusters.

Analysis: The RMSE distribution often inversely correlates with the stability seen in the NRMSE plot. High consumption clusters may naturally exhibit higher RMSE values even if their patterns are relatively predictable. However, the alignment of high RMSE with high NRMSE in certain clusters points to "problematic"profiles that are both high-magnitude and irregular. The variability in RMSE confirms the necessity of the clustering approach: a global model would likely be skewed by the high-magnitude errors of a few volatile clusters, whereas segmented modeling allows for more targeted strategies.

6.6 Cluster Prediction Analysis

The following analysis evaluates the forecasting performance across distinct consumer profiles. By comparing the structural characteristics of the clusters with their respective prediction results, we can isolate the specific behaviors that lead to model failure or success.

6.6.1 Cluster 1

Cluster 1 represents a segment of consumers characterized by significant irregularity. As shown in the Global RMSE analysis, this cluster exhibits a Normalized RMSE (NRMSE) greater than 1.0, indicating that the error variance exceeds the variance of the data itself.

Performance Analysis: The prediction graph (Figure 44) highlights the limitations of historical autocorrelation for this group. The consumption at time t shows little correlation with $t - 168$ (one week prior), causing the baseline model to miss the timing of peaks completely. The Random Forest model, while attempting to smooth the signal, ultimately fails to predict the magnitude of the spikes, suggesting that this cluster is driven by external, non-periodic events.

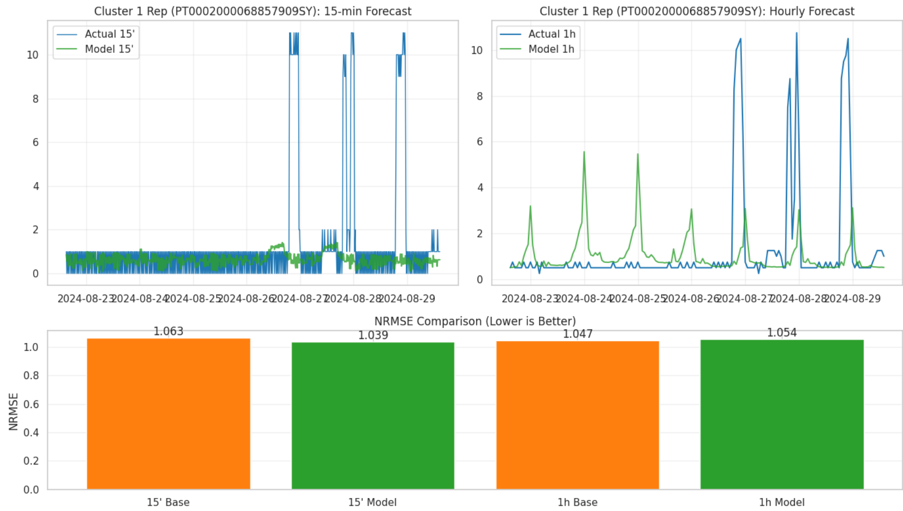


Figura 43: Cluster 1: Consumption Profile. The profile shows erratic peaks without a clear weekly cycle.

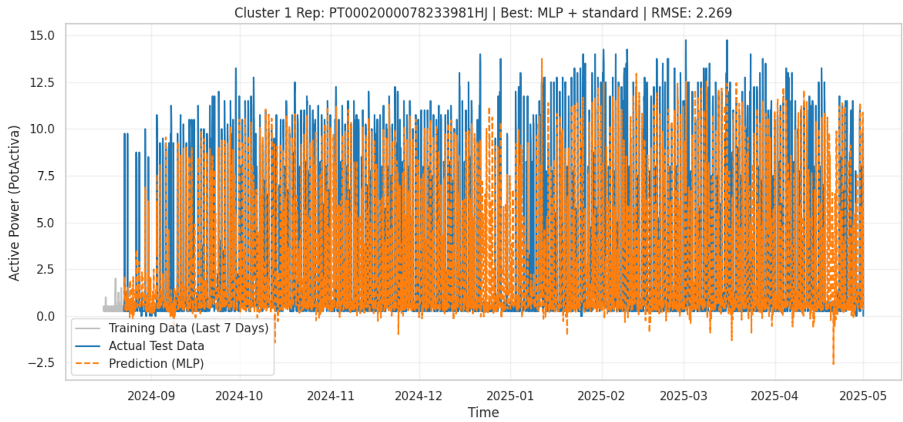


Figura 44: Cluster 1: Prediction vs Actual. The Weekly Persistence Baseline (Green) fails to capture the stochastic spikes, leading to high residuals.

6.6.2 Cluster 6

Cluster 6 was identified during the quality checks as a particularly problematic group, often containing "erroneous"or extremely noisy data points. It represents the "worst-case"scenario for forecasting.

Performance Analysis: Figure 46 demonstrates a near failure of all modeling approaches. The massive residuals indicate that the consumption patterns in Cluster 6 are effectively random or governed by variables not present in the dataset. This confirms the findings from the silhouette analysis, which flagged Cluster 6 as a small, non-compact cluster that should potentially be treated as outliers rather than a valid consumer segment.

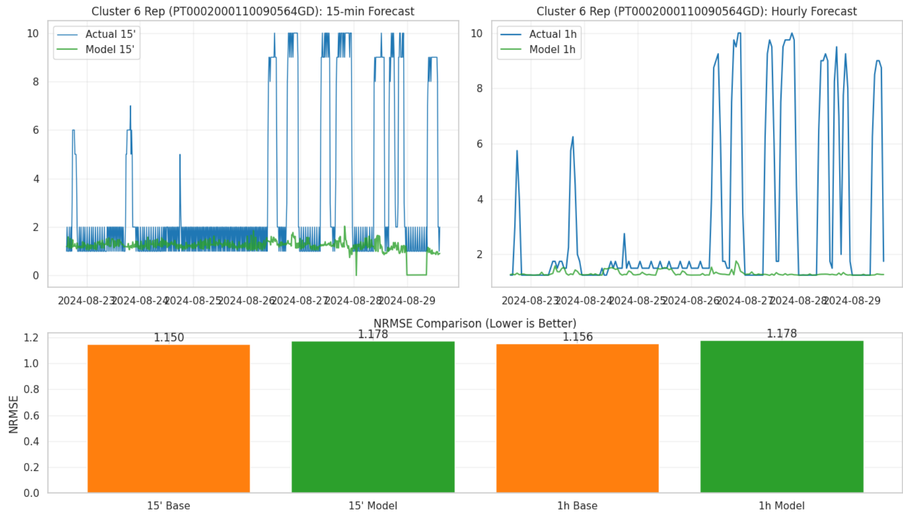


Figura 45: Cluster 6: Irregular Profile. The profile exhibits chaotic fluctuations.

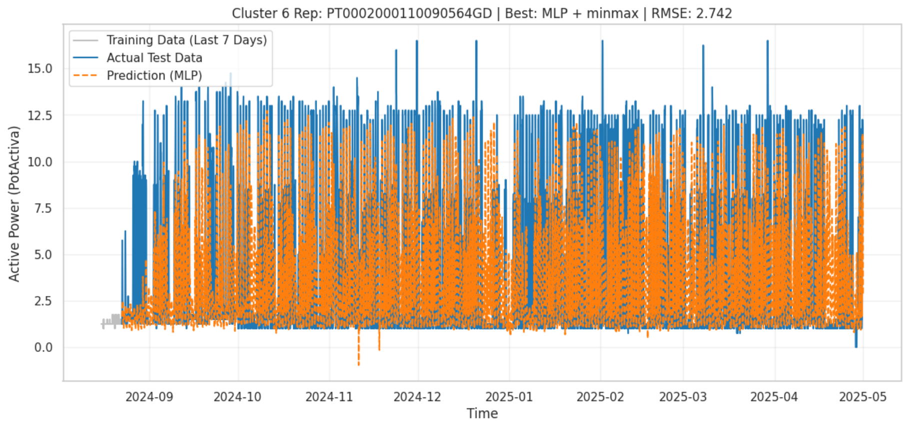


Figura 46: Cluster 6: Prediction vs Actual. The massive divergence between the prediction and actual values confirms the unpredictable nature of this segment.

6.6.3 Cluster 7

Cluster 7 represents a group of consumers with negligible or near-zero electricity consumption throughout the analyzed period. As observed in the Global RMSE analysis, this cluster achieves a Normalized RMSE (NRMSE) equal to zero, reflecting the absence of variability in the underlying load signal rather than superior forecasting capability.

Performance Analysis: The prediction results shown in Figure 48 indicate perfect alignment between predicted and observed values for both the baseline and the trained model. This outcome is not driven by learned temporal patterns, but rather by the trivial nature of the forecasting task, as the consumption signal exhibits no meaningful temporal variation.

In this context, the weekly persistence assumption holds exactly, and machine learning models offer no additional benefit over the baseline. Clusters such as Cluster 7 can therefore artificially improve aggregate error metrics if not treated separately, highlighting the importance of cluster-

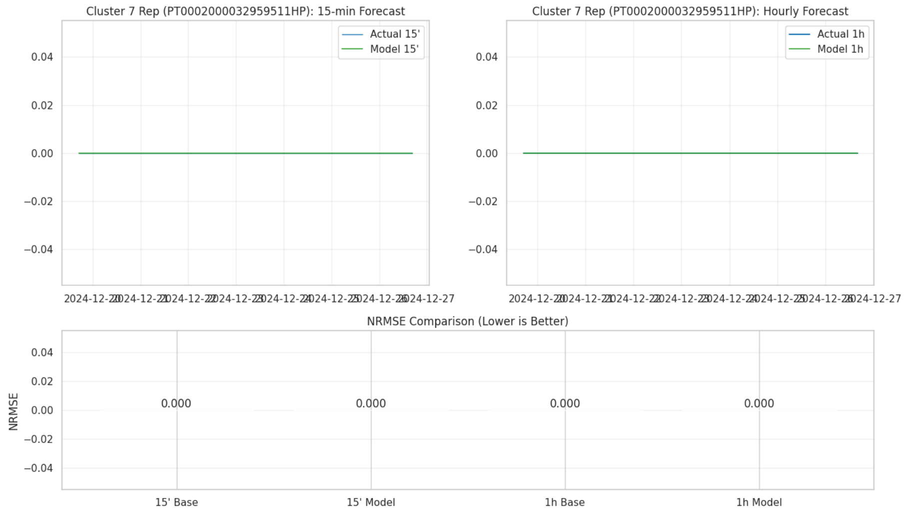


Figura 47: Cluster 7: Consumption Profile. The profile remains flat at (or very close to) zero, indicating inactive or quasi-inactive electricity usage.

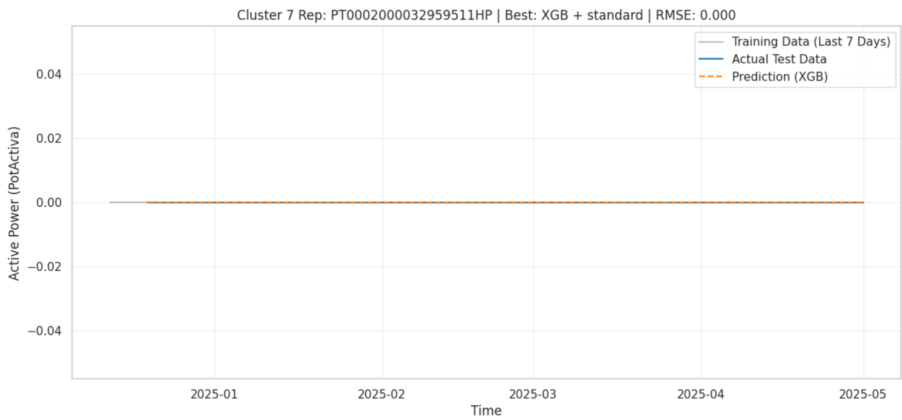


Figura 48: Cluster 7: Prediction vs Actual. Both the Weekly Persistence Baseline and the XGBoost model perfectly match the observed load due to the absence of consumption dynamics.

aware performance interpretation and the careful handling of near-zero variance consumers in forecasting studies.

6.6.4 Cluster 10

In sharp contrast to the previous groups, Cluster 10 exemplifies the "ideal"consumer for time-series forecasting, characterized by a structured and repetitive schedule.
Performance Analysis: The success of the simple models here is evident. As seen in Figure 50, the Weekly Persistence Baseline (Green) overlaps almost perfectly with the actual consumption (Black). The low RMSE and NRMSE values for this cluster confirm that when a consumer follows a routine schedule (likely commercial/industrial working hours), complex Machine Learning models offer little to no advantage over simple lag-based heuristics.

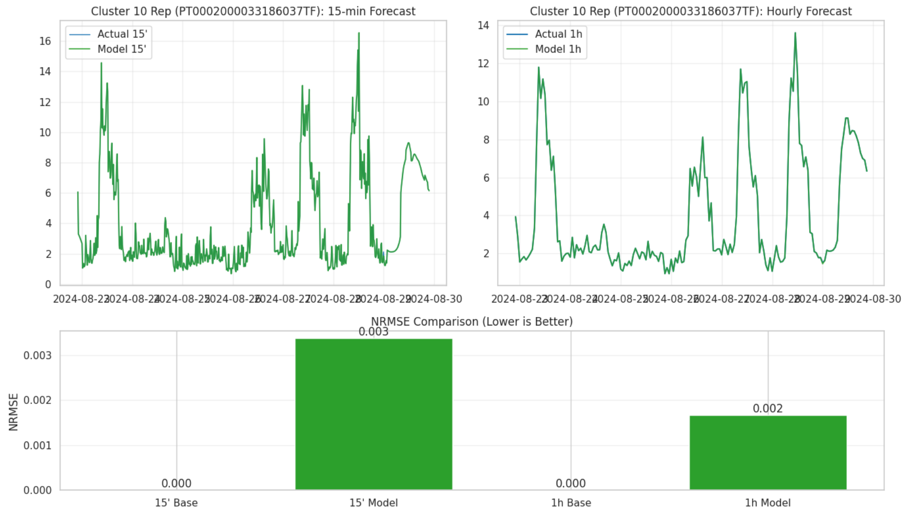


Figura 49: Cluster 10: Structured Profile. Clear and consistent baselines.

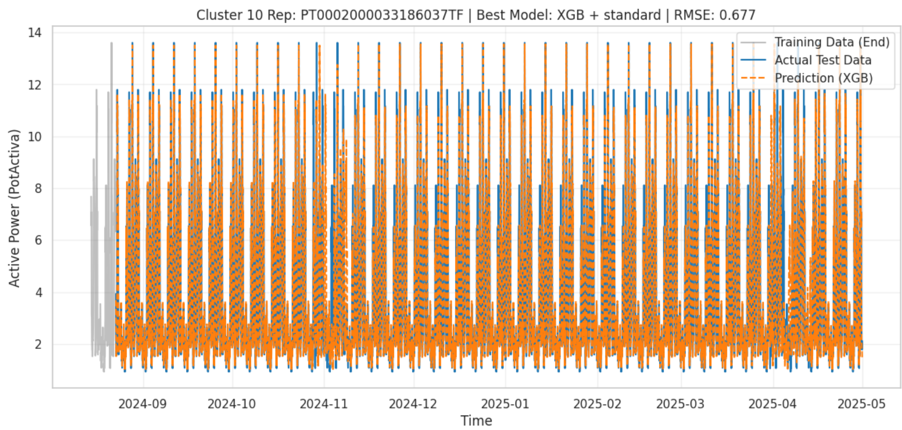


Figura 50: Cluster 10: Prediction vs Actual. The Weekly Persistence Baseline tracks the actual consumption with high precision.

6.7 Summary of Comparative Performance

A comparative analysis of the selected clusters reinforces the central finding of this study: forecast accuracy is strongly determined by the degree of regularity in the underlying consumption profile. As consumer behavior transitions from highly irregular to strictly repetitive or inactive, the forecasting task correspondingly shifts from complex to trivial.

- **High Volatility (Clusters 1 & 6):** These clusters are characterized by erratic consumption patterns with weak temporal autocorrelation. Accurate forecasting in this regime requires the incorporation of exogenous information or contextual variables, as models relying solely on historical load fail to anticipate the timing and magnitude of demand fluctuations.
- **Near-Zero Consumption (Cluster 7):** This cluster comprises inactive or quasi-inactive consumers whose load profiles exhibit negligible variance. As a result, both baseline and

machine learning models attain perfect or near-perfect accuracy. Such outcomes reflect the trivial nature of the forecasting task rather than genuine model skill, underscoring the need to identify and treat near-zero consumption profiles separately to avoid biasing aggregate performance metrics.

- **High Stability (Cluster 10):** Strong and consistent seasonal patterns dominate the load profiles in this cluster. Consequently, simple persistence-based baselines achieve performance comparable to more complex models, offering a computationally efficient and robust forecasting solution.

7 Final Conclusion

This project investigated the characterization and forecasting of electricity consumption patterns for individual CPEs of the Municipality of Maia, following the CRISP-DM methodology and combining unsupervised and supervised learning techniques. Starting from high-frequency energy measurements, the study systematically explored data preparation, feature engineering, consumer segmentation, and forecasting model evaluation.

A comprehensive data understanding and preparation phase revealed strong sparsity, high variability, and significant heterogeneity across CPEs. Temporal gaps and missing values were addressed through structured interpolation at the 15-minute level, enabling consistent downstream analysis. Exploratory analysis confirmed the presence of dominant weekly seasonality, frequent near-zero consumption periods, and highly skewed distributions, motivating both feature-based characterization and consumer segmentation.

In Exercise 1, clustering techniques (K-means and DBSCAN) were applied to engineered features capturing temporal behavior, load variability, and reactive power contributions. The resulting clusters effectively separated CPEs with strictly periodic, semi-regular, volatile, near-zero, and highly irregular consumption profiles. Correlation analysis and feature selection ensured that the retained feature set was informative and non-redundant. Cluster interpretation highlighted that consumption regularity, rather than absolute magnitude, is the key factor differentiating consumer behavior.

Exercise 2 demonstrated that forecasting performance is strongly driven by the statistical properties of the consumption profiles. A simple weekly persistence baseline proved to be a remarkably strong benchmark, often outperforming more complex models such as ARIMA, Random Forest, XGBoost, and MLP, particularly for clusters exhibiting stable and repetitive patterns. Temporal aggregation from 15-minute to hourly resolution consistently reduced prediction error, confirming that high-frequency noise limits the effectiveness of complex models when strong weekly seasonality dominates.

In Experiment 3, a cluster-aware modeling strategy was introduced to address the limitations of global forecasting approaches. By combining cluster-specific normalization, target transformations, and model selection, the analysis showed that no single preprocessing pipeline or regression model is optimal across all consumer types. Tree-based models demonstrated robustness to feature scaling, while linear and distance-based models required careful normalization. Seasonal target transformations were particularly effective for highly regular clusters, whereas logarithmic transformations improved performance for spiky and high-variance profiles.

Overall, the results confirm that electricity consumption forecasting at the individual CPE level is inherently heterogeneous. Forecast accuracy depends more on behavioral regularity than on model complexity, and cluster-aware strategies are essential for robust and interpretable forecasting. For highly regular consumers, simple persistence-based baselines remain the most reliable and efficient solution, while irregular and volatile consumers require additional contextual or exogenous information to achieve meaningful improvements.

This work highlights the importance of combining unsupervised segmentation with supervised forecasting and demonstrates that tailoring modeling choices to consumer behavior leads to more effective and realistic energy forecasting pipelines. Future work should focus on incorporating external variables (e.g., weather conditions, calendar events, or operational schedules) and extending the framework to probabilistic forecasting in order to better capture uncertainty in highly irregular consumption profiles.